

3.6 Carboxylic Acids & Derivatives

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.6 Carboxylic Acids & Derivatives
Difficulty	Hard

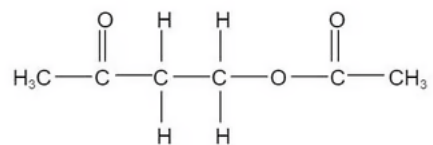
Time allowed: 20

Score: /10

Percentage: /100

Question 1

In the presence of an H^+ catalyst, compound X reacts with ethanoic acid to produce the compound below.



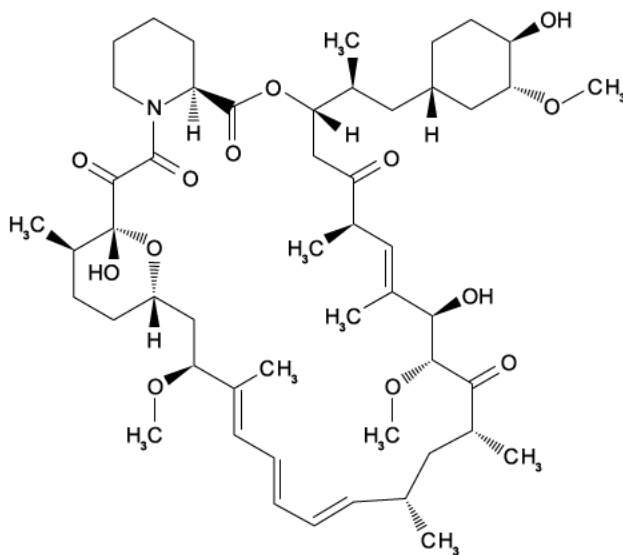
What is the molecular formula of compound X?

- A. $\text{C}_4\text{H}_8\text{O}$
- B. $\text{C}_4\text{H}_8\text{O}_2$
- C. $\text{C}_2\text{H}_6\text{O}_2$
- D. $\text{C}_2\text{H}_6\text{O}_3$

[1 mark]

Question 2

Sirolimus is a drug used to treat patients after kidney transplants.

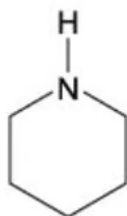


Sirolimus

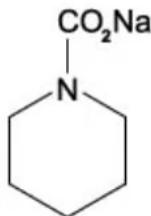
Sirolimus produces an equimolar mixture of two organic products, on reaction with hot aqueous sodium hydroxide.

What is the structural formula of the product with the lower relative molecular mass?

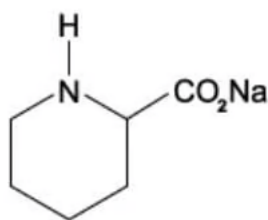
A.



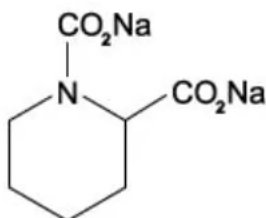
B.



C.



D.



[1 mark]

Question 3

Compound **K**, C₅H₁₂O, is oxidised by acidified sodium dichromate (VI) to compound **L**.

Compound **L** reacts with butan-2-ol in the presence of a little concentrated sulfuric acid to give liquid **M**.

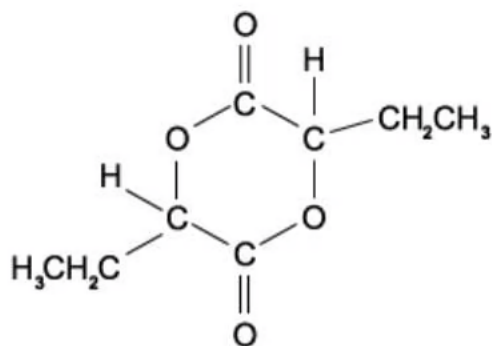
What could be the formula of liquid **M**?

- A. (CH₃)₂CHCH₂CO₂C(CH₃)₃
- B. CH₃(CH₂)₃CO₂(CH₂)₃CH₃
- C. CH₃(CH₂)₃CO₂CH(CH₃)CH₂CH₃
- D. CH₃(CH₂)₂CO₂CH₂CH₂(CH₃)₂

[1 mark]

Question 4

Compound **X** can be made from 2-hydroxybutanoic acid.



Compound **X**

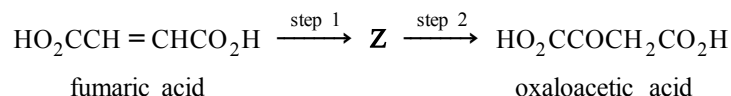
What should be heated with 2-hydroxybutanoic acid in order to make compound **X**?

- A. Aluminium oxide
- B. Concentrated sulfuric acid
- C. Dilute sodium hydroxide
- D. Acidified potassium manganate under reflux

[1 mark]

Question 5

Oxaloacetic acid can be synthesised from fumaric acid in a two-step process involving the intermediate **Z**.



Each step involved can be achieved in the laboratory by a single reagent.

What could be the intermediate **Z** and the reagent for step 2?

	Z	Reagent for step 2
A	$\text{HO}_2\text{CCH}(\text{OH})\text{CH}_2\text{CO}_2\text{H}$	Fehling's solution
B	$\text{HO}_2\text{CCH}(\text{OH})\text{CH}_2\text{CO}_2\text{H}$	Warm acidified $\text{K}_2\text{Cr}_2\text{O}_7$
C	$\text{HO}_2\text{CCHBrCH}_2\text{CO}_2\text{H}$	Warm acidified KMnO_4
D	$\text{HO}_2\text{CCHBrCH}(\text{OH})\text{CO}_2\text{H}$	Warm NaOH (aq)

[1 mark]

Question 6

Use of the Periodic Table is relevant to this question.

During a lesson, a teacher heated ethyl propanoate under reflux with aqueous sodium hydroxide. The two organic products are then separated, purified and weighed.

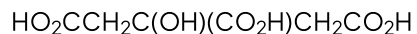
Out of the total mass of products obtained, what is the percentage by mass of each product?

- A. 50.0 % and 50.0 %
- B. 57.7 % and 42.3 %
- C. 61.7 % and 38.3 %
- D. 67.6 % and 32.4 %

[1 mark]

Question 7

Citric acid gives lemons and other citrus fruits their sour taste.



citric acid

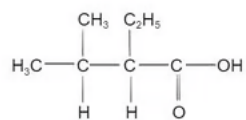
What is the volume of 0.2 mol dm^{-3} sodium hydroxide solution required to neutralise a solution containing 0.008 mol of citric acid?

- A. 40 dm^3
- B. 80 dm^3
- C. 120 dm^3
- D. 160 dm^3

[1 mark]

Question 8

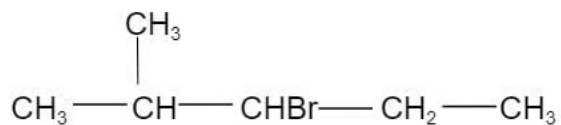
The compound 2-ethyl-3-methylbutanoic acid gives rum its signature smell.



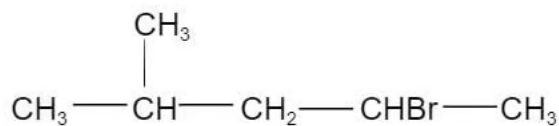
2-ethyl-3-methylbutanoic acid

Which compound will produce 2-ethyl-3-methylbutanoic acid by heating under reflux with alcoholic sodium cyanide and subsequent acid hydrolysis of the reaction product?

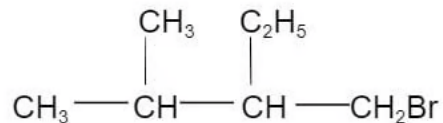
A.



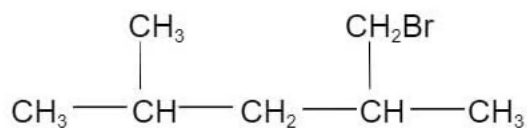
B.



C.



D.



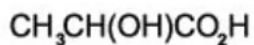
[1 mark]

Question 9

Digestion of various foods and medicines can cause acidic compounds to pass through the body. The stomach wall can become sensitive to acidic compounds, resulting in stomach ulcers.

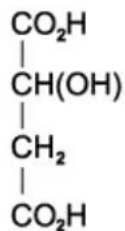
Which of the following is the most acidic compound?

A.



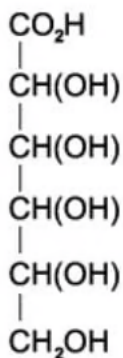
lactic acid

B.



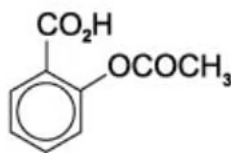
malic acid

C.



gluconic acid

D.



aspirin

[1 mark]

Question 10

Use of the Period Table is relevant to this question.

30 g of an alcohol and 30 g of a carboxylic acid are heated under reflux together to produce 33 g of propyl butanoate.

What is the yield of the ester?

- A. 25 %
- B. 38 %
- C. 50 %
- D. 66 %

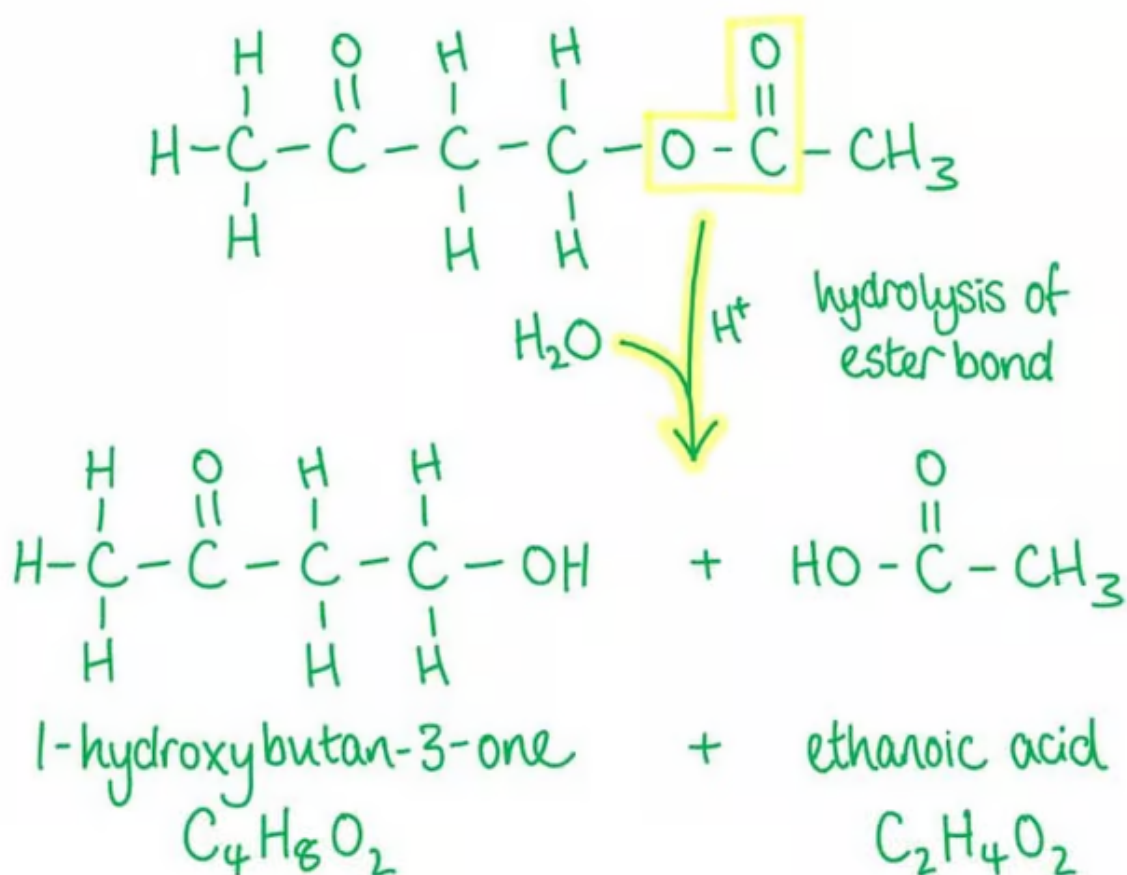
[1 mark]

Model Answers: Hard

1

The correct answer is **B** because:

- Ethanoic acid is a 2-carbon carboxylic acid that will react with an alcohol to produce the ester in the question.

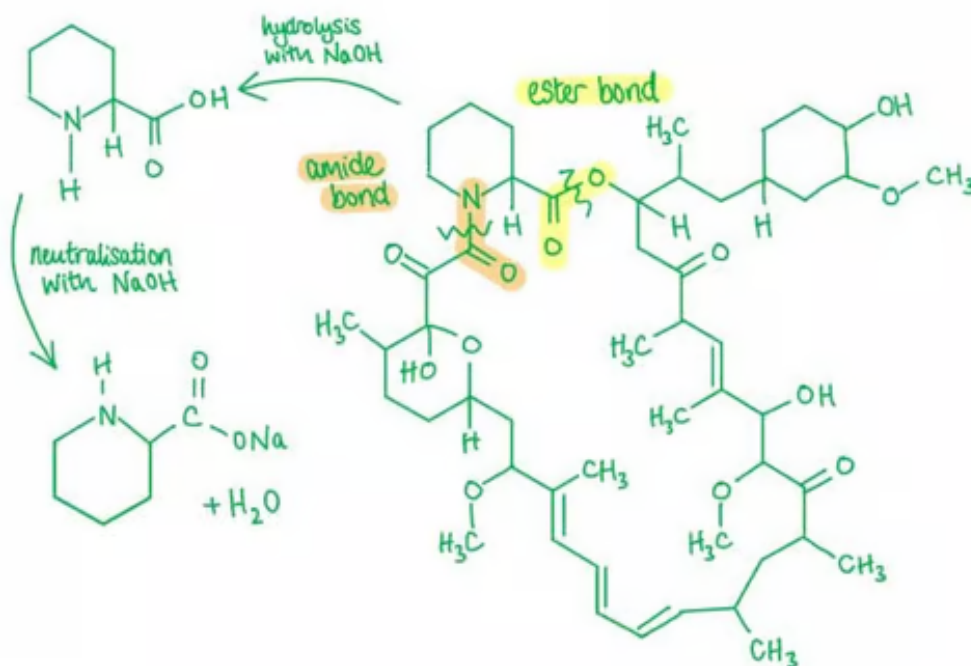


2

The correct answer is **C** because:

- Sodium hydroxide, NaOH , first catalyses the hydrolysis of the ester and amide bond.
- Then, the carboxylic acid produced undergoes a neutralisation reaction with

sodium hydroxide to produce a salt ($\text{-RCOO}^-\text{Na}^+$) and water.

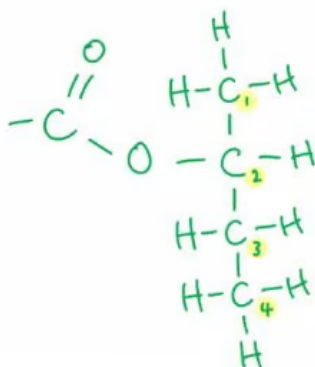


Amide bonds are not specifically in the syllabus, but they are very similar to ester bonds, the amine group acts similarly to alcohol oxygen in an ester and both bonds can be broken with NaOH. They appear in proteins, linking amino acids together within proteins.

3

The correct answer is **C** because:

- The molecular formula of compound **K** suggests it is either a ketone or an aldehyde.
- As compound **K** can be **oxidised** it must be an **aldehyde**, making compound **L** a **carboxylic acid**.
- When a carboxylic acid reacts with an **alcohol** with a **sulfuric acid catalyst** an **ester** is produced.
- The latter half of the ester formula is from the alcohol therefore you have to look for $\text{OCH}(\text{CH}_3)\text{CH}_2\text{CH}_3$ which is seen in option **C**: $\text{CH}_3(\text{CH}_2)_3\text{CO}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$



A is incorrect as this would be the product if compound **L** reacted with 2-methylpropan-2-ol.

B is incorrect as this would be the product if compound **L** reacted with butan-1-ol.

D is incorrect as this would be the product if compound **L** reacted with 2-methylpropan-1-ol.

4

The correct answer is **B** because:

- 2-hydroxybutanoic acid has both an alcohol and carboxylic acid group and therefore 2 molecules can react together in a condensation reaction, forming a diester (a molecule with two ester groups).
- Concentrated sulfuric acid acts as a catalyst in condensation reactions that result in esters.

A is incorrect as **aluminium oxide** (Al_2O_3) acts as a **catalyst** in the **dehydration** of **alcohols to alkenes**, therefore reacting 2-hydroxybutanoic acid with aluminium oxide would produce but-2-en-oic acid $\text{CH}_3\text{CH}=\text{CHCOOH}$

C is incorrect as 2-hydroxybutanoic acid and **sodium hydroxide** would undergo a **neutralisation reaction** resulting in an **organic salt** $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{COONa}$ and water.

D is incorrect as reacting 2-hydroxybutanoic acid with **acidified potassium manganate** under reflux would cause the **alcohol group** to **oxidise** to first an aldehyde and then a **carboxylic acid**, resulting in butanedioic acid ($\text{HOOCCH}_2\text{CH}_2\text{COOH}$)

5

The correct answer is **B** because:

- Fumaric acid has an **alkene** functional group and oxaloacetic acid has a **carbonyl** functional group.
- Therefore, in step 1 the **alkene** group in fumaric acid is **hydrated** to produce an alcohol (**Z**) and in step 2 this alcohol group is **oxidised** to a **ketone**.

alcohol (**Z**) and in step 2 this alcohol group is **oxidised** to a **ketone**.

- The structure of **Z** is $\text{HO}_2\text{CCH}(\text{OH})\text{CH}_2\text{CO}_2\text{H}$ as there is an alcohol group on the **2nd carbon** (which has the alkene group in fumaric acid).
- The reagent of step 2 is the oxidising agent **acidified $\text{K}_2\text{Cr}_2\text{O}_7$** which is heated.

A is incorrect as Fehling's solution is an oxidising agent that converts aldehydes into carboxylic acids, in this reaction blue CuSO_4 is reduced to red Cu_2O which can be used to show the presences of aldehydes. Fehling's solution does not oxidise alcohols because CuSO_4 is not a strong enough oxidising agent.

C is incorrect as $\text{HO}_2\text{CCHBrCH}_2\text{CO}_2\text{H}$ cannot be turned into oxaloacetic acid in 1 step using the oxidizing agent KMnO_4 .

D is incorrect as $\text{HO}_2\text{CCHBrCH}(\text{OH})\text{CO}_2\text{H}$ would require more than one step to synthesise and its reaction with NaOH would result in a neutralization reaction between NaOH and the carboxylic acid groups to produce the salt $\text{NaO}_2\text{CCHBrCH}(\text{OH})\text{CO}_2\text{Na}$

6

The correct answer is **C** because:

- Heating the ester ethyl propanoate under reflux with aqueous sodium hydroxide as a catalyst will produce ethanol and propanoic acid.
- **Calculate the molecular weights of each product:**
 - Ethanol:
 - $[2 \times 12 (\text{carbon})] + [1 \times 16 (\text{oxygen})] + [6 \times 1 (\text{hydrogen})] = \mathbf{46 \text{ g/mol}}$
 - Propanoic acid:
 - $[3 \times 12 (\text{carbon})] + [2 \times 16 (\text{oxygen})] + [6 \times 1 (\text{hydrogen})] = \mathbf{74 \text{ g/mol}}$
- **Percentage by mass of each product:**
 - Total product weight = $46 + 74 = 120$
 - % ethanol = $\frac{46}{120} \times 100\% = \mathbf{38.3\%}$
 - % propanoic acid = $\frac{74}{120} \times 100\% = \mathbf{61.7\%}$

7

The correct answer is **C** because:

• **Calculate moles NaOH required:**

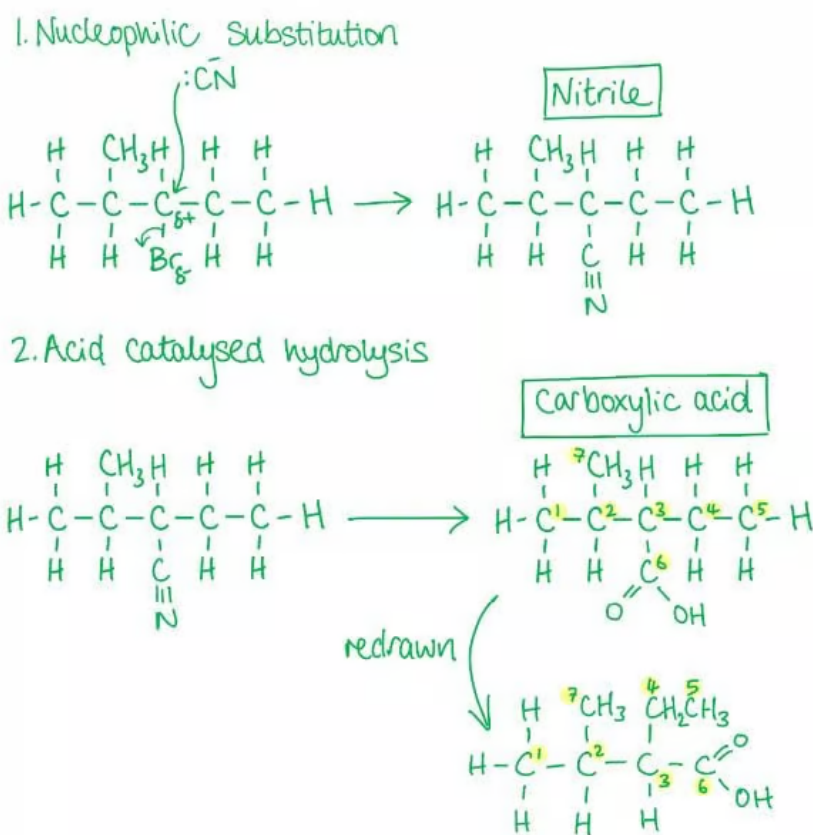
- Citric acid contains 3 carboxylic acid groups, each will be neutralised by a sodium hydroxide, NaOH molecule.
- Therefore, for every 1 moles of citric acid 3 moles of NaOH is required.
- 0.008 moles citric acid is used.
- $0.008 \times 3 = 0.024$ moles NaOH is required.

• **Calculate volume of NaOH required by rearranging $\text{moles} = \text{conc} \times \text{vol}$**

- $\frac{\text{moles (mol)}}{\text{concentration (mol dm}^{-3})} \times 1000 = \text{volume (dm}^3)$
- $\frac{0.024 \text{ mol}}{0.2 \text{ mol dm}^{-3}} \times 1000 = 120 \text{ dm}^3$

8

The correct answer is **A** because:



9

The correct answer is **B** because:

- The most **acidic** compound is the compound that will increase the **hydrogen ion concentration** the most, this means the one that **dissociates** with the most H^+ .
- In these organic acids, the most acidic will be the one with the most carboxylic acid groups.
- **Malic acid** has **2 carboxylic acid groups** and is therefore the most acidic.

A, C & D are incorrect as lactic acid, gluconic acid and aspirin only have 1 carboxylic acid group.

10

The correct answer is **D** because:

- 1 mole of alcohol reacts with 1 mole of carboxylic acid to produce 1 mole of ester.
- Propyl butanoate is produced from propanol and butanoic acid.
- **Known quantities:**
 - Mass of propanol; 30g
 - Molecular mass of propanol; 60 g mol^{-1} *from periodic table*
 - Mass of butanoic acid; 30g
 - Molecular mass of butanoic acid; 88 g mol^{-1} *from periodic table*
 - Mass of propyl butanoate produced; 33g
 - Molecular mass of propyl butanoate; 146 g mol^{-1} *from periodic table*
- **Calculate the number of moles of each reactant:**
 - $\text{moles} = \frac{\text{mass}}{M_r}$
 - $\text{moles propanol} = \frac{30}{60} = 0.50 \text{ mol}$
 - $\text{moles butanoic acid} = \frac{30}{88} = 0.34 \text{ mol}$
 - *Therefore, butanoic acid is the limiting reactant – only 0.34 mol of each react.*
- **Calculate theoretical yield of propyl butanoate:**

- $\text{moles} \times M_r = \text{mass}$
- $0.34 \text{ mol} \times 146 \text{ g mol}^{-1} = 49.8 \text{ g}$

- **Calculate percentage yield of propyl butanoate:**

- $\% \text{ yield} = \frac{\text{actual yeild}}{\text{theoretical yeild}} \times 100\%$
- $\% \text{ yield} = 33.0 \text{ g} / 49.8 \text{ g} \times 100\%$
- $\% \text{ yield} = 66.26 \% = \mathbf{66 \%}$